

## **ABSTRACT**

In today's digital world, it has become increasingly difficult to distinguish between real images and AI-generated or manipulated images. This creates serious concerns related to misinformation, fake content, and lack of trust in digital media. Our project aims to solve this problem by developing an AI-based image recognition system that can accurately identify whether a given image is real or AI-generated.

The system uses advanced machine learning and deep learning techniques to analyze patterns, textures, and hidden features within an image. Unlike traditional methods, our solution not only classifies the image but also provides a confidence score, which represents the probability of the image being real or AI-generated (for example, AI-generated: 96%, Real: 4%). This helps users understand how reliable the prediction is.

Additionally, the system includes an image manipulation detection feature. This enhancement allows the model to detect whether the image has undergone modifications such as compression, filtering, resizing, or editing. By identifying these alterations, the system provides deeper insights into the authenticity and integrity of the image.

The proposed solution can be implemented as a web-based application where users can upload an image and instantly receive results. This project can be useful in various domains such as social media verification, journalism, cybersecurity, digital forensics, and content moderation.

Overall, our project enhances trust in digital content by combining image classification, probability scoring, and manipulation detection into a single, user-friendly platform.